

Late Breaking Results: Power Rail Routing for Advanced Multi-Layered Printed Circuit Boards

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ABSTRACT

This paper proposes a power rail routing flow for advanced multi-layered printed circuit boards (PCBs) to optimize segment area and via usage while satisfying IR drop requirements. With increasing current/voltage demands in modern PCBs, ultra-wide power rails may consume most routing space and cause significant routing problems. We present an effective overlap-aware rail sizing technique to distribute routing spaces appropriately according to current/voltage demands and a resistance-aware A*-search algorithm to resolve overlapping regions by rail detouring. Experimental results show that our work significantly outperforms the state-of-the-art rail router in the metal area and runtime, achieving respective reductions of 49% and 28%, without any current/voltage violations.

1 INTRODUCTION

Power distribution networks (PDNs) are designed to deliver power to printed circuit board (PCB) components while satisfying power integrity (PI) requirements. Specifically, the source-to-drain voltage difference must be kept within a certain level. To meet this requirement in high-speed, high-density electronics nowadays, layout engineers have encountered several challenges. First, each board-level rail delivers power with a specific current/voltage demand. In addition, with strict voltage drop constraints and large current demands, a PDN usually requires parallel rail segments on multiple layers to deliver sufficient current. Most importantly, the power rails may consume significant space and cause routing problems. Sizing any of them may directly affect others' shapes and thus resistance, as shown in Figure 1(a). Due to these challenges, PDN designs often undergo time-consuming manual refinements. With increasing design complexity, an automatic routing flow is needed to optimize the PDN designs while satisfying PI constraints.

Automatic power planning studies can be divided into two levels: (1) *Chip Level* and (2) *Board Level*. Researchers focused on stripe sizing for chip-level power planning to deliver sufficient power to standard cells or macros. Chowdhury *et al.* [1] and Tan *et al.* [2] minimized the stripe area by adjusting the current and voltage of a PDN circuit in alternating orders. However, stripe overlaps and multiple current/voltage demands are not considered in chip-level PDN design, so their methods cannot readily apply to board-level applications. Not much work deals with board-level power rail routing. Bairamkulov *et al.* [3] proposed a grid-based power rail routing scheme to minimize rail impedance with given area constraints. However, they routed the power rails greedily with unspecified orders, which may lead to routability issues in complex cases. In addition, they considered only a single layer without parallel segments for each rail, which may not be sufficient for current-hungry, high-demanding drains.

To remedy these drawbacks, we propose a power rail routing flow for multi-layered PCBs, where routing spaces can be distributed according to the current/voltage demands of each rail. The design flow consists of two main techniques: (1) an overlap-aware sizing technique that optimizes the segment width and via usage, and (2) a resistance-aware A*-routing algorithm that optimizes the segment path. These techniques achieve better segment area and runtime.

2 PROBLEM FORMULATION

Given each rail's source/drain ports, each port's feasible region for inserting vias (defined as a via region), each source port's voltage/current supply, each drain port's voltage/current demand, and the obstacles on each layer, generate the metal shape of rail segments and insert vias in the via regions such that the segment area and the via usage are minimized without segment overlaps while the current/voltage demand of each drain port is satisfied.

3 ALGORITHMS

Figure 1(b) shows our algorithm flow, where the four stages are detailed in the following subsections.

3.1 Preprocessing

This stage generates a grid map for each layer and a multi-layered topology for each rail. In the grid map, each grid is considered a basic unit for routing and via insertion. Thus, the grid width is set as the via pad diameter, making each grid a

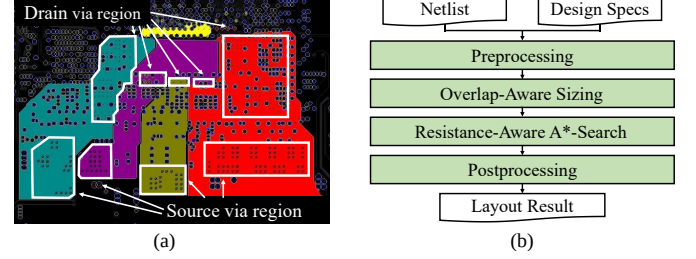


Figure 1: (a) An example of a PDN in one layer of a multi-layered PCB (from ASUS). (b) Our algorithm flow.

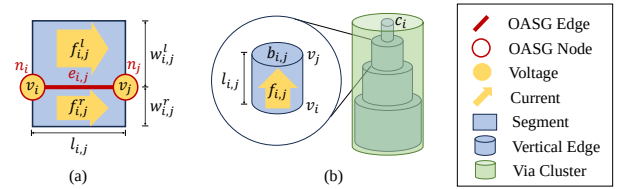


Figure 2: (a) Transformation from a horizontal edge to a rail segment. (b) Transformation from a stack of vertical edges to a via cluster.

legal candidate for inserting a single via. On the other hand, a rail topology stores all the current delivery path candidates. To generate rail topologies without overlapping with obstacles, we first construct an obstacle-aware spanning graph (OASG) [4] that connects the source and drains on each layer. Then, OASGs of adjacent layers are connected with vertical edges that represent via clusters. A via cluster is defined as a set of vias inserted into the same via region. These OASGs and vertical edges form a multi-layered rail topology.

3.2 Overlap-Aware Sizing

This section presents mathematical programming to determine the width of each rectangular segment and the cross-sectional area of each via cluster. Figure 2 illustrates the construction of rail segments and via clusters from a rail topology. For each horizontal edge $e_{i,j}$, the corresponding rail segment takes $e_{i,j}$ as its spine and forms a rectangular metal area according to the left and right widths ($w_{i,j}^l$, $w_{i,j}^r$), given as follows:

$$w_{i,j}^r = \frac{\rho l_{i,j} f_{i,j}^r}{h_{i,j} (v_i - v_j)}, \quad (1)$$

where ρ denotes the resistivity constant, $l_{i,j}$ the length of $e_{i,j}$, $h_{i,j}$ the layer thickness of $e_{i,j}$, $f_{i,j}^r$ the current flowing on the right of $e_{i,j}$, and v_i the voltage on a node n_i . The left segment width can be expressed in a similar form.

For each vertical edge $e_{i,j}$, its required cross-sectional area $b_{i,j}$ is calculated similarly. Since plated-through-hole vias have the same area on all layers, a via cluster should have the largest required area of all its vertical edges.

In our mathematical formulation, our objective is to minimize the total segment area and via usage:

$$\min \quad \alpha \sum_{e_{i,j} \in EH} a_{i,j} + \beta \sum_{z_i \in Z} c_i + \gamma \sum_{e_{i,j} \in EH} d_{i,j}, \quad (2)$$

where $a_{i,j}$ denotes the segment area of a horizontal edge $e_{i,j}$, c_i the cross-sectional area of the via cluster z_i , $d_{i,j}$ the difference between the left and right segment areas of the edge $e_{i,j}$, and α , β , and γ are user-defined parameters. All these terms are uniquely defined with fixed currents $f_{i,j}^r$, $f_{i,j}^l$ and voltages v_i , which can be derived from Equation (1). The third term encourages even distribution of segment widths between the left and right parts of an edge. This strategy reduces the peripheral current delivery distance and thus reduces the error of our estimated resistance.

To consider the spatial competition among different rail segments, we restrict the widths of horizontal edges to avoid overlaps. Figure 3 shows an example of this segment non-overlapping constraint, formulated as follows:

$$w_{i,j}^r \times \cos \theta_1 + w_{k,m}^l \times \cos \theta_2 \leq \hat{w}_q, \quad \forall q \in Q^D, \quad (3)$$

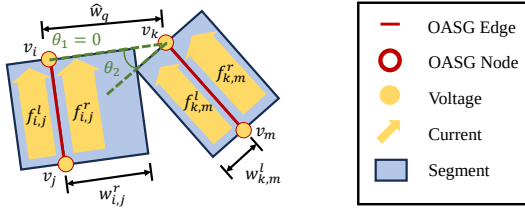


Figure 3: Illustration of the non-overlapping constraint.

where Q^D denotes the set of 2-edged non-overlapping constraints, $\hat{w}_q$ the minimum distance between $e_{i,j}$ and $e_{k,m}$, θ_1 the angle between $e_{i,j}$ and the shortest line segment, and θ_2 is defined similarly for $e_{k,m}$ in the non-overlapping constraint q . We also impose similar constraints on each single edge to avoid overlapping with obstacles.

Besides the segment non-overlapping constraint, we also add three sets of constraints to produce feasible circuits. First, according to Kirchhoff's Law, we restrict the entering and leaving currents of a node to be the same. Secondly, we restrict each edge's current and voltage drop to share the same direction. Last but not least, we restrict the cross-sectional area of each via cluster to be under an upper bound value, determined by the number of grids covered by the corresponding via region.

Inspired by Chowdhury's [1] and Tan et al.'s [2] approaches, we split the main problem into current-based and voltage-based sizing subproblems and solve them in alternating orders. The current-based subproblem fixes the voltages and minimizes the objective by adjusting edge currents, which is solved with linear programming (LP). In contrast, the voltage-based subproblem fixes the currents and minimizes the objective by adjusting nodal voltages, which is solved with a sequence of linear programming (SLP) to speed up the convex optimization process. Since the main problem is convex, the alternating LP and SLP process can be proved to obtain the optimal solution once the objective value converges.

Segment overlaps may exist in high-current cases, which can be solved by rail detouring in later stages. We use Lagrangian relaxation on the segment non-overlapping constraints to minimize segment overlaps.

3.3 Resistance-Aware A*-Search

We develop an A*-search algorithm to route rail segments with their previously assigned widths on the constructed grid map. Our A*-search cost is designed below:

$$c(g_i) = c_l(g_i) + c_t(g_i) + c_c(g_i) + c_h(g_i), \quad (4)$$

where g_i is the current grid, c_l is the X-architecture path length, c_t is the shortest distance from g_i to the source-target line segment, c_c is the congestion cost, and c_h is the estimated X-architecture distance from g_i to the target grid. Since our heuristic cost never overestimates the actual cost, the algorithm can be proven to find an optimal solution if exists [5].

Our goal is to maintain the resistance of each segment, decided in Section 3.2. However, the segment width grows with its path length under fixed resistance, as shown in Equation (1). Since the exact path length is unknown during A*-search, we may underestimate the segment width and congestion during routing. To remedy this problem, we design the congestion cost as an expected value. For example, the congestion cost of stepping upward can be calculated as follows:

$$c_c(g_i) = \sum_{w=\hat{w}}^{\hat{w}+\Delta w} p(w) \sum_{x=x_i-\lfloor w/2 \rfloor}^{x_i+\lfloor w/2 \rfloor} m\left(x, y_i + \left\lfloor \frac{w}{2} \right\rfloor\right), \quad (5)$$

where $p(w)$ is the probability of the exact segment width w , $\hat{w}$ is the segment width decided in the previous sizing stage, and $m(x, y)$ is the congestion on the grid located at a coordinate (x, y) . $p(w)$ is designed to consider the probability distribution of the detoured length. Since longer detoured paths exist with smaller possibilities, $p(w)$ is the highest when $w = \hat{w}$, and soon decays with growing w . The congestion cost of stepping to other X-architecture directions can be calculated similarly.

We route the rail segments by performing A*-search and updating congestion iteratively. First, we apply A*-search to obtain the optimal path and update the exact routing width to be proportional to the path length. With the updated width, we retrace the path to occupy the grids as the metal region of the segment. Finally, we complete the routing by updating the grid map congestion. With this resistance-aware A*-search algorithm, we can route the rails while maintaining each segment's resistance as decided previously with a global view.

3.4 Postprocessing

In Postprocessing, we insert vias into via regions and refine the shapes of power rails. When inserting vias into a via region, we first calculate the number of vias k from the cross-sectional area of the via cluster decided in Section 3.2. We then evenly select k grids from those covered by a via region and insert a single via in each of these

Table 1: Comparison of the segment area and runtime for SPROUT [3] and our design flow.

Benchmark	Power Supply		Segment Area (grid)		Runtime(s)	
	Current (A)	Voltage (V)	SPROUT*	Ours	SPROUT*	Ours
Case 1	110	1.65	218434	77894	977	416
Case 2	80 ~ 220	1.5 ~ 1.65	83767	70827	218	220
Case 3	35 ~ 110	1.5 ~ 1.65	147403	70312	375	221
Case 4	35 ~ 94	1.5 ~ 1.65	121392	68102	307	582
Comp.	-	-	1.9665	1	1.3909	1

grids. Since the grid width equals the pad diameter of a via, our insertion guarantees legal spacing among all vias.

In the refinement stage, we resolve the overlapping grids and further minimize the segment area while satisfying the current/voltage requirements. We first construct a multi-layered partial element equivalent circuit (PEEC) [6] for each rail and adopt nodal analysis [7] to get the current and voltage of each grid. A grid overlapped by multiple rails is assigned to the rail on which the grid delivers the most current (normalized by the current demand). We also modify the SmartGrow method [3] to add the grids with higher currents and remove the grids with lower currents for each rail. This modified method considers the grids in all layers concurrently and keeps only the grids with higher currents. Thus, the segment area can be further minimized while satisfying the current/voltage requirements.

4 EXPERIMENTAL RESULTS

We implemented our rail routing algorithm in the C++ programming language and conducted the experiments on a 2.9GHz AMD Ryzen Threadripper 3990X 128-Core processor with 128GB memory. The mathematical programming and PEEC-based simulation were solved with the Gurobi Optimizer [8] and Eigen Library [9], respectively. Our benchmarks were modified from the industrial cases, where the current demands ranged from 35 to 220 amperes, and the maximum acceptable voltage drop was 0.05 volts for each rail. The high current features reveal the excellent spatial distribution ability of our proposed method.

To evaluate the effectiveness and efficiency of our proposed method, we compared our results with the state-of-the-art rail router, SPROUT [3]. For fair comparisons, we re-implemented SPROUT with the same grid size (0.4064 millimeters). However, SPROUT could not meet the high current demands because it did not consider parallel current delivery paths on multiple layers. Consequently, we extended SPROUT to handle these cases. We first executed SPROUT layer by layer. Besides, SPROUT inserted insufficient vias because it did not consider current demands in inter-layered connections. Thus, we added the maximum number of vias to favor SPROUT's operations and did not compare via usage to avoid unfair comparison.

Table 1 summarizes the resulting segment area and runtime of SPROUT and our proposed method. On average, our work achieves a 49% reduction in segment area and a 28% reduction in runtime. In addition, all current/voltage demands were satisfied by both methods. The results show that our proposed rail router distributes routing spaces more effectively and efficiently.

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